

An excerpt from Kan – sei No.95

【Regarding to points of attention with guidance on municipal solid waste treatment activity】

Promulgated date : November 4, 1977 (昭和 52 年 11 月 4 日)

Kan – sei No.95(環整 95 号)

Revision: February 1, 1990 (平成 2 年 2 月 1 日) Ei-kan No.22(衛環 22 号)

(The letter to Municipal Solid Waste Treatment Bureau in each prefecture by Director, Environment Improvement Division, Water Supply and Environmental Sanitation Department, Environmental Health Bureau)

Regarding to enhancement of guidance on municipal solid waste treatment activity

The letter by Director, Environment Improvement Division, Water Supply and Environmental Sanitation Department, Ministry of Health and Welfare (Kan-sei No. 95 as of November 4, 1997)

Garbage quality analysis

(An excerpt from Kan-sei No.95)

1. Collecting garbage

(1) Collecting garbage from collecting lorry

Collect at least 10 kg of garbage per 1 collecting lorry, which is chosen at random, until at least 200 kg are collected in total.

(2) Collecting garbage from garbage pit

Collect garbage more than 200 kg after mixing it enough in garbage pit.

2. Sample preparation

First, mix collected sample with such as a shovel on the dry floor such as concrete, take out content from bagged garbage and subdivide very large garbage. Next, collect garbage between 5 kg and 10kg as sample to reduce it in several times with quartering method while mixing.

Notes

About large waste (such as blanket, tire, bamboo tree and oil tin), it is desirable to add them to sample to shred them at the end after taking them out from waste in advance.

For example, in case to reduce waste with quartering method 4 times, if taking out blanket from waste after the second time, it is added quarter of that blanket to sample because it is due to be reduced in quarter by 2 more time reduction.

3. Measurement and analysis

(1) Unit volume weight

Put sample in Article 2 into a known capacity container and drop it from about 30 cm in height 3 times. Put only loss capacity of sample more into container if capacity is reduced by it.

Unit volume weight (or apparent specific gravity) is calculated on the basis of a following equation (1).

$$\text{Unit volume weight (kg/m}^3\text{)} = \text{Sample weight [kg]} / \text{Container capacity [m}^3\text{]} \cdot \cdot \cdot (1)$$

(2) Moisture

After weight sample in Article 3 (1), drying it up at 105°C±5°C until obtaining constant weight, and then weight sample. Moisture is calculated on basis of a following equation (2)

$$\text{Moisture(\%)} = ((\text{Weight before drying [kg]} - \text{Weight after drying [kg]}) / \text{Weight before drying [kg]}) \times 100(\%) \cdot \cdot \cdot (2)$$

(3) Garbage types component analysis

Calculate weight ratio (%) to weight each component as following 6 components are standards after putting total amount of sample used in Article 3 (2) on such as plastic sheet.

1. Paper, Blankets and Comforters
2. Plastic, Synthetic resin, Rubber, Leathers
3. Wood, Bamboo, Straw
4. Kitchen garbage (including animal and plant residues, eggshell, shell)
5. Incombustible waste
6. Other (waste passed through 5mm pore size sieve)

(4) Ash content

Pulverize each component under 2 mm by operating pulverizer except incombustible waste of 6 separated components in Article 3(3). Put a part of it into crucible and heat it up at 105°C ± 5°C for 2 hours.

After weighting this, heat it up at 800°C for 2 hours and weight it.

Ash content is calculated on basis of following equations (3), (4) and (5).

$$\text{Each component ash content (\%)} = (\text{Weight after heating up [kg]} / \text{Weight before heating up [kg]}) \times 100 \cdot \cdot \cdot (3)$$

$$\text{Dried garbage ash content (\%)} = \frac{\sum_{i=1}^6 (A_i \cdot B_i)}{\sum_{i=1}^6 A_i} \cdot \cdot \cdot (4)$$

A_i: Weight ratio of each component i calculated on (3) (%)

B_i: Ash content of each component i (%)

(It is that incombustible waste ash content is 100.)

$$\text{Raw garbage ash content (\%)} = \text{Dried garbage ash content (\%)} \times ((100 - \text{Moisture(\%)})/100) \cdot \cdot \cdot (5)$$

(5) Combustible component

Combustible component is calculated on a following equation (6).

$$\text{Combustible component (\%)} = 100 - \text{Moisture(\%)} - \text{Raw garbage ash component} \cdot \cdot \cdot (6)$$

(6) Lower calorific value

It is possible to estimate raw garbage lower calorific value on a following equation (7).

$$HI = 4,500V - 600W \cdot \cdot \cdot (7)$$

HI: raw garbage lower calorific value (kcal/kg)

V: Raw garbage combustible content (%)

W: Raw garbage moisture content (%)

4. Guidance for garbage quality analysis

- (1) It shall be carried out immediately to collect and reduce sample.
- (2) Preserve sample in a closed container to prevent moisture content change in sample if weighting sample before drying to measure moisture content is carried out on a different day from sampling.
- (3) Provide for safety to set an operation supervisor to prevent accidents such as fall if sampling is carried out beside pit.
- (4) Take care not to receive injuries in contact with raw garbage directly such as reducing and separating garbage. Take appropriate emergency measures such as sterilization if someone should receive injuries.

Three components in garbage

One of important indexes is three components in garbage. Obtain “Moisture”, “Ash content”, “Combustible content” through the process such as drying up, separation and burning as figure” Three component in garbage”.

$$\text{Raw garbage} = \text{Moisture content} + \text{Combustible content} + \text{Ash content}$$

Moisture content

Loss in weight by drying is moisture content. Drying temperature is 105°C. Generally wet weight is denominator in garbage analysis although there is an example dry weight is denominator in the field except garbage infrequently. Moisture content is never over 100 % in that case, but moisture content over 100 % is a possibility when dry weight is denominator.

Moisture content (%)=(Wet weight – Dry weight)/Wet weight×100

Combustible content

Separate combustible from incombustible by separating dried garbage. Actually obtain category 2 weight by calculation to separate in many items more commonly.

Additionally combustible can be separated burned content and unburned content. Burned content is comestible.

Comestible content (%) = Raw garbage – Moisture - Ash

Component items

Actually convert with calculating by burning a small amount of “Incombustible”. This process is “Ash measurement”.

Ash content

“The residue of burned garbage” by burning “Combustible” (This is also said “Ash content”) plus “Incombustible” is “Ash content”.

Add “Incombustible” to Ash content by thinking it unburns 100 % when it is burned.

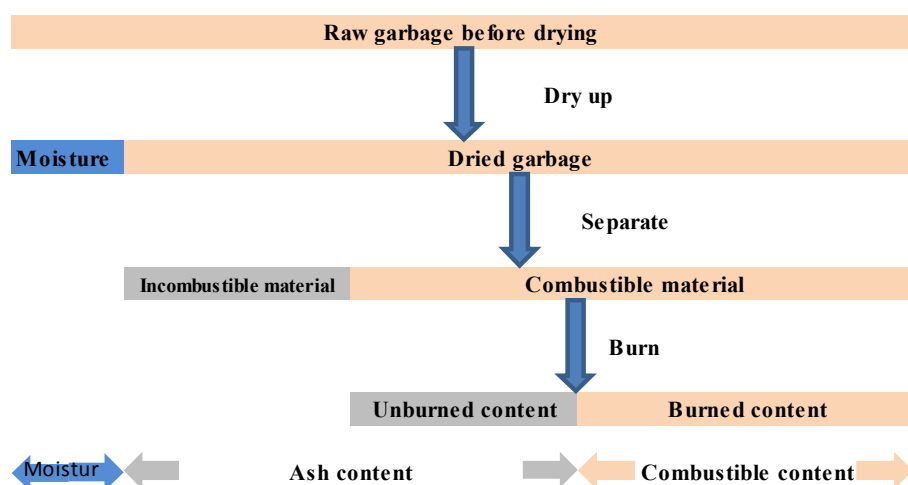


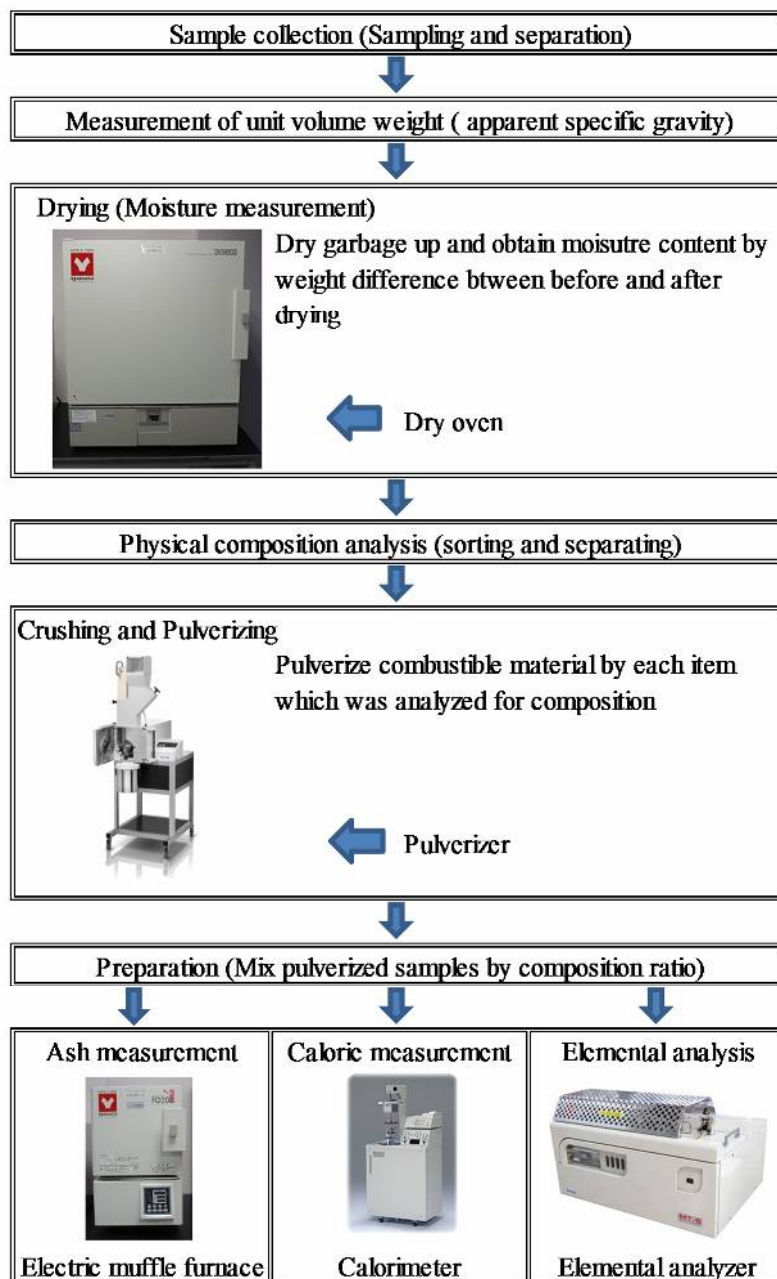
Figure-1. Three components in Garbage

- Obtain moisture content by drying garbage up.
- Separate incombustible.
- Obtain unburned content to burn a part of combustible. Convert it to all combustible.
- Incombustible and unburned content of combustible equivalent is ash content(as total).
- What subtracts moisture and ash from total is combustible content. It is expressed as a percentage generally.

Simplified example

- When drying up weight of 10 kg raw garbage is 5 kg,
 $(10-5)/10 \times 100 = 50\%$ Moisture content
- When Incombustible weight is 1 kg by separating dried garbage, combustible is 4 kg.
- When ash ratio to dried combustible 4 kg is 10 %, total ash content is
Incombustible (1 kg) + Dried combustible ash content ($4 \text{ kg} \times 10 \%$)
- Weight ratio to 10 kg raw garbage (ash ratio) is $1.4/10 \times 100 = 14 \%$
- $100 - \text{Moisture content (50)} - \text{Ash content (14)} = \text{Combustible content (36\%)}$

Garbage analysis process



* Elemental analysis items

Carbon, Hydrogen, Nitrogen, Sulfur, Chlorine, Oxygen (calculation) and so on.

Circular cone quartering method

Normally use circular cone quartering method to obtain typical samples with less deviation.

Circular cone quartering method based on “Kan-sei No.95

■ Obtain more than 200 kg samples from garbage collecting lorry or garbage pit.

* It can have possibility to have deviation at this time. Try to obtain typical sample.

It is necessary to be agitated enough in obtaining samples from pit. Judge adoption when there is a deviation in obvious and start over in some cases.

■ Uniformize samples to make big-sized materials smaller (separate materials not to handle) and take out hollow bag.



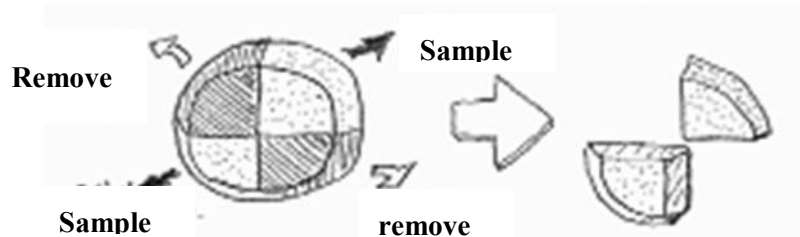
■ Tier a cone of garbage to drop from an upper point.



■ Smoothen garbage to push down from the top downward



■ Choose two across part of garbage with random to quarter and it should be a sample.



■ Obtain samples between 5 kg and 10 kg and repeat the steps above for 2 or 3 times